Branch Coverage Report

Project: Railway Reservation System (RRS), Team 13 Jackfruit Phase-1
Generated: 12-Nov-2026
Environment: Python 3.11.9, Windows
Tool: coverage.py (branch measurement on)

Summary

Branches	Taken both ways	Branch coverage
406	309	76.1%

A branch is one outcome of a decision (for example the `if` taken and the `if` skipped). Each decision has at least two.

Branch coverage per file

File	Branches	Covered	Partial lines	Branch %
app.py	32	1	1	3.1%
db.py	2	0	0	0.0%
modules/admin.py	12	11	1	91.7%
modules/allocation.py	8	7	1	87.5%
modules/audit.py	10	8	2	80.0%
modules/auth.py	24	23	1	95.8%
modules/booking.py	18	16	2	88.9%
modules/cancellation.py	14	12	2	85.7%
modules/monitoring.py	16	11	5	68.8%
modules/ops_routes.py	4	4	0	100.0%
modules/payment.py	10	9	1	90.0%
modules/pnr.py	24	19	3	79.2%
modules/pnr_routes.py	16	13	1	81.2%
modules/promotion.py	8	8	0	100.0%
modules/rbac.py	8	8	0	100.0%
modules/refund.py	4	4	0	100.0%
modules/search.py	16	15	1	93.8%
modules/seat_allocation.py	14	13	1	92.9%
modules/session_guard.py	10	9	1	90.0%
modules/train_admin.py	62	58	4	93.5%
modules/train_admin_routes.py	10	7	3	70.0%
modules/transport_security.py	8	8	0	100.0%
reliability.py	4	3	1	75.0%
repository/admin_repo.py	16	13	3	81.2%
repository/audit_repo.py	2	2	0	100.0%
repository/booking_repo.py	6	4	2	66.7%
repository/cancellation_repo.py	2	0	0	0.0%
repository/payment_repo.py	16	15	1	93.8%
repository/promotion_repo.py	10	8	2	80.0%
repository/train_admin_repo.py	20	0	0	0.0%

Branches never taken (lowest files first)

- `db.py` (0.0%): line->target 35->36, 35->41
- `repository/cancellation_repo.py` (0.0%): line->target 42->43, 42->59
- `repository/train_admin_repo.py` (0.0%): line->target 65->66, 65->67, 77->78, 77->79, 85->86, 85->89, 98->99, 98->100, 105->106, 105->111, 122->123, 122->124, 124->125, 124->143, 131->132, 131->136, 150->151, 150->152, 153->154, 153->155

- `app.py` (3.1%): line->target 42->43, 43->44, 43->45, 54->55, 54->56, 61->62, 61->75, 67->68, 67->72, 68->69, 68->70, 80->81, 80->109, 92->93, 92->96, 104->105, 104->107, 134->135, 134->141, 138->139, 138->141, 165->166, 165->170, 170->171, 170->175 (+6 more)
- `repository/booking_repo.py` (66.7%): line->target 35->36, 83->84
- `modules/monitoring.py` (68.8%): line->target 51->53, 66->68, 75->76, 105->106, 112->113
- `modules/train_admin_routes.py` (70.0%): line->target 63->64, 78->79, 87->88
- `reliability.py` (75.0%): line->target 11->exit
- `modules/pnr.py` (79.2%): line->target 13->15, 40->47, 42->44, 45->40, 45->46
- `modules/audit.py` (80.0%): line->target 16->18, 30->31

Explanation of gaps

Repository files contain SQL that needs a live Supabase database. Their logic is tested with a fake database cursor, so some SQL-only lines run only in manual system testing (see System_Validation_Report.md).

Full line-by-line view: `docs/reports/coverage_html/index.html` (open in a browser; red lines were never run, yellow lines had a branch not taken).